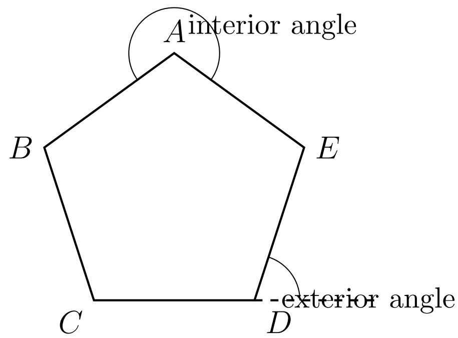
Interior and Exterior Polygon Angle Sums

Interior angle sums, a missing angle inside a polygon, exterior angle sums, and the interior and exterior angles of a regular polygon.

- Every polygon here is convex. Let n be the number of sides.
- State the theorem you are using before you substitute.
- Give angle answers in degrees; give side-count answers as a whole number.



How angles are named here. No values are shown: use the numbers given in each problem. An *interior* angle sits inside the polygon at a vertex; its *exterior* angle is formed by extending one side, and the two together form a straight line.

1. Find the sum of the interior angles of a convex hexagon.

Answer: ________________ degrees

2. Find the sum of the interior angles of a convex 15-gon.

Answer: ________________ degrees

3. A regular octagon has all of its interior angles equal. Find the measure of one interior angle.

Answer: ________________ degrees

4. A regular 12-gon has all of its exterior angles equal. Find the measure of one exterior angle.

Answer: ________________ degrees

5. Four of the interior angles of a convex pentagon measure $102°$, $118°$, $96°$, and $131°$. Find the measure x of the fifth interior angle.

Answer: ________________ degrees

6. Each exterior angle of a regular polygon measures $24°$. How many sides n does the polygon have?

Answer: ________________

7. Each interior angle of a regular polygon measures $150°$. How many sides n does the polygon have? Set up the equation from the interior angle sum rather than guessing.

Answer: ________________

8. The four interior angles of a convex quadrilateral are in the ratio $1 : 2 : 3 : 4$, so they can be written x, $2x$, $3x$, and $4x$. Find x.

Answer: ________________ degrees

9. The interior angles of a convex polygon add to $1980°$. How many sides n does the polygon have?

Answer: ________________

10. **Justify.** Prove or explain in complete sentences why the exterior angles of *every* convex polygon add to $360°$, no matter how many sides it has.
Use the interior angle sum $(n-2)180°$ and the fact that an interior angle and its exterior angle at the same vertex form a linear pair. Your explanation must work for a general n, not just for one example.